

5.1 Energy: Conservation, Work, Power & Efficiency

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.1 Energy: Conservation, Work, Power & Efficiency
Difficulty	Medium

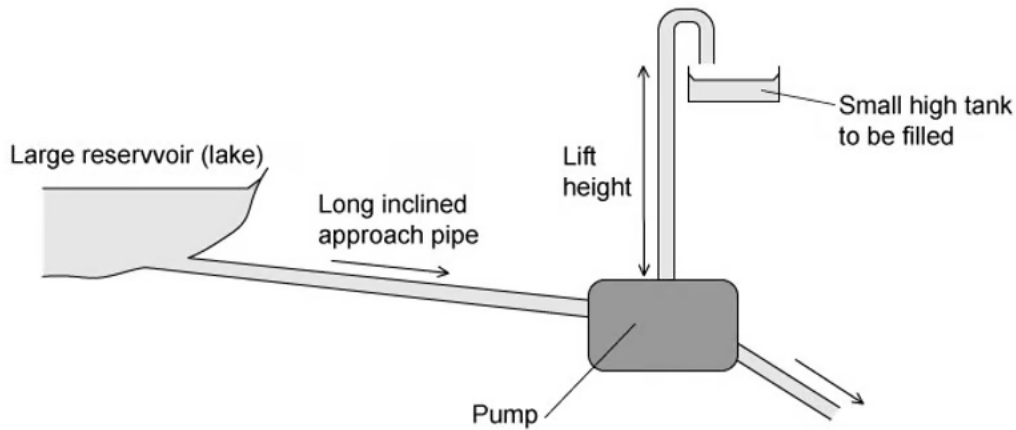
Time allowed: 20

Score: /13

Percentage: /100

Question 1

The diagram shows a pump called a hydraulic ram



In one such pump the long approach pipe holds 500 kg of water. A valve shuts when the speed of this water reaches 2.0 m s^{-1} and the kinetic energy of this water is used to lift a small quantity of water by a height of 15 m.

The efficiency of the pump is 10 %.

Which mass of water could be lifted 15 m?

- A. 0.15 kg
- B. 0.68 kg
- C. 1.5 kg
- D. 6.8 kg

[1 mark]

Question 2

Trains supply coal to a power station. The table below gives quantities describing the operation of the power station.

	Symbol	Unit
power station output	P	W
number of trains per day	N	
mass of coal on a train	M	kg
energy from 1 kg of coal	J	J
number of seconds in one day	S	s

Which expression gives efficiency of the power station?

A. $\frac{PS}{NMJ}$

B. $\frac{PSN}{MJ}$

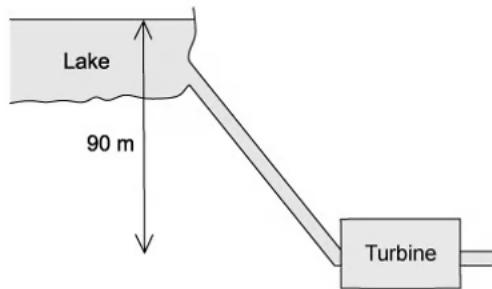
C. $\frac{NMJ}{PS}$

D. $\frac{NM}{PSJ}$

[1 mark]

Question 3

Water flows from a lake into a turbine that is a vertical distance of 90 m below the lake, as shown



The mass flow rate of the water is 2400 kg min^{-1} . The turbine has an efficiency of 75 %.

What is the output power of the turbine?

- A. 26 kW
- B. 35 kW
- C. 1.6 MW
- D. 2.1 MW

[1 mark]

Question 4

Initially, four identical uniform blocks, each of mass m and thickness h , are spread on a table.



How much work is done on the blocks in stacking them on top of one another?

- A. $3 mgh$
- B. $6 mgh$
- C. $8 mgh$
- D. $10 mgh$

[1 mark]

Question 5

The first column in the table gives four examples of work being done. The second column gives more detail of the action.

Which row is **not** correct?

	Example	Detail
A	a girl dives from a diving board into a swimming pool	work is done by the girl against gravity as she falls
B	a man pushes a car along a level road	work is done by the man against friction
C	an electron is accelerated towards a positively-charged plate	work is done on the electron by the electric field of the plate
D	a piston is pushed outwards as a gas expands	work is done on the atmosphere by the gas

[1 mark]

Question 6

The data below are taken from a test of a petrol engine for a motor car.

power output	150 kW
fuel consumption	20 litres per hour
energy content of fuel	40 MJ per litre

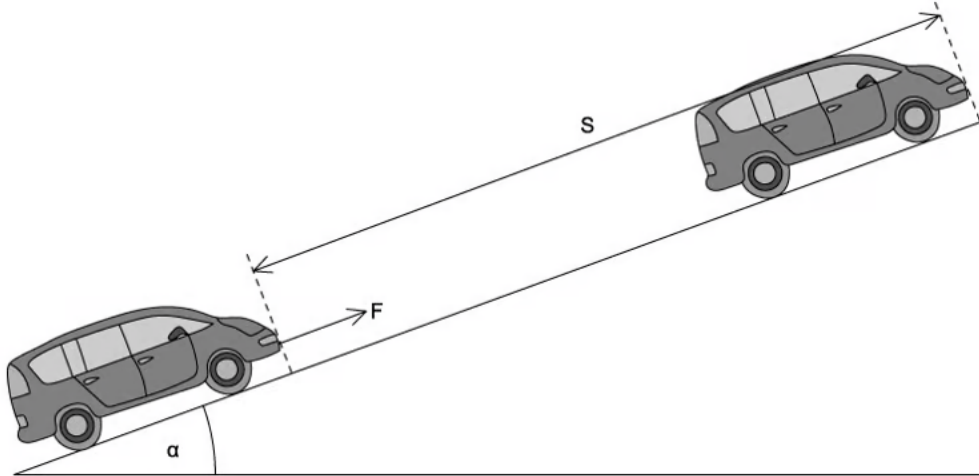
Which expression will evaluate the efficiency of the engine?

- A. $\frac{150 \times 10^3}{40 \times 10^6 \times 20 \times 60 \times 60}$
- B. $\frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$
- C. $\frac{150 \times 10^3 \times 40 \times 10^6 \times 20}{60 \times 60}$
- D. $\frac{150 \times 10^3 \times 20}{40 \times 10^6 \times 60 \times 60}$

[1 mark]

Question 7

A constant force F , acting on a car of mass m , moves the car up a slope through a distance s at constant velocity v . The angle of the slope to the horizontal is α .



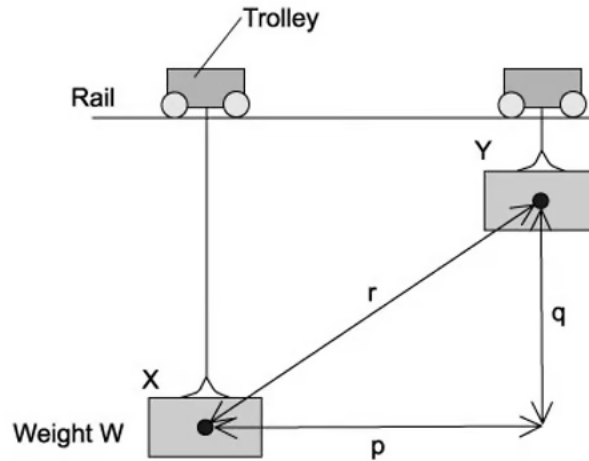
Which expression gives the efficiency of the process?

- A. $\frac{mgs \sin \alpha}{Fv}$
- B. $\frac{mv}{Fs}$
- C. $\frac{mv^2}{2Fs}$
- D. $\frac{mg \sin \alpha}{F}$

[1 mark]

Question 8

A weight W hangs from a trolley that runs along a rail. The trolley moves horizontally through a distance p and simultaneously raises the weight through a height q .



As a result, the weight moves through a distance r from X to Y . It starts and finishes at rest.

How much work is done on the weight during this process?

- A. Wp
- B. $W(p + q)$
- C. Wq
- D. Wr

[1 mark]

Question 9

A bungee jumper has 24 kJ of gravitational potential energy at the top of his jump. He is attached to an elastic rope which starts to stretch after a short time of free fall. Assume that energy loss through air resistance is negligible.

	GPE / kJ	EPE / kJ	KE / kJ
Top	24	0	0
Bottom	0	24	0

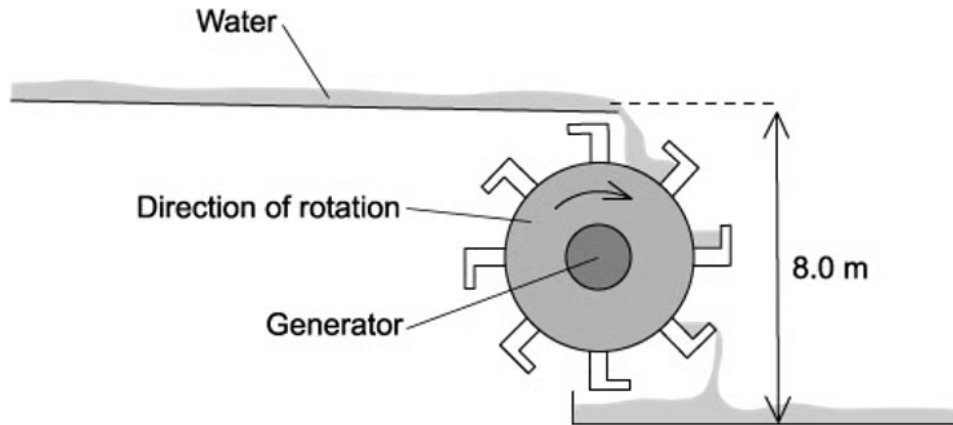
What are the possible values when the jumper is half-way down?

	GPE / kJ	EPE / kJ	KE / kJ
A	12	10	2
B	12	8	4
C	8	8	8
D	12	2	10

[1 mark]

Question 10

The diagram shows the design of a water wheel which drives a generator to produce electrical energy. The flow rate of the water is 200 kg s^{-1} . The generator supplies a current of 32 A at a voltage of 230 V .



Ignoring any changes in kinetic energy of the water, what is the efficiency of the system?

- A. 14 %
- B. 16 %
- C. 22 %
- D. 47 %

[1 mark]

Question 11

A conveyor belt is driven at velocity V by a motor. Sand drops vertically on to the belt at a rate of $m \text{ kg s}^{-1}$.

What is the additional power needed to keep the conveyor belt moving at a steady speed when the sand starts to fall on it?

- A. $\frac{1}{2}mv$
- B. mv
- C. $\frac{1}{2}mv^2$
- D. mv^2

[1 mark]

Question 12

The force resisting the motion of a car is taken as being proportional to the square of the car's speed. The magnitude of the force at a speed of 20 m s^{-1} is 800 N.

What effective power is required from the car's engine to maintain a steady speed of 40 m s^{-1} ?

- A. 32 kW
- B. 64 kW
- C. 128 kW
- D. 512 kW

[1 mark]

Question 13

A mass is raised vertically. In a time t , the increase in its gravitational potential energy is E_p and the increase in its kinetic energy is E_k .

What is the average power input to the mass?

- A. $(E_p - E_k)t$
- B. $(E_p + E_k)t$
- C. $\frac{E_p - E_k}{t}$
- D. $\frac{E_p + E_k}{t}$

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- In this question we need to consider conservation of energy since energy is being transformed from one form into another
- The kinetic energy of the water is being converted into gravitational potential energy of the water
- However, since the efficiency of the pump is only 10%, only 10% of the kinetic energy is converted into GPE
- 10% of KE of the water = GPE of water lifted
 - $0.1 \times \frac{1}{2}mv^2 = m_w gh$
- Where m_w is the mass of the water lifted
 - $0.1 \times \frac{1}{2} \times 500 \times 2^2 = m_w \times 9.81 \times 15$
 - $m_w = \frac{0.1 \times 500 \times 4}{2 \times 9.81 \times 15} = 0.68 \text{ kg}$
- Therefore, 0.68 kg of water could be lifted by 15 m

2

The correct answer is **A** because:

- Efficiency is a quantity that describes a ratio of two similar appropriate quantities, so, the numerator and the denominator must have the same units
- In this scenario, we can give the efficiency in terms of input and output energy per day, we will need to use the following equations:
- $\text{Efficiency} = \frac{\text{output energy per day}}{\text{total energy available per day}}$
- $\text{Power} = \frac{\text{energy}}{\text{time}}$
- $\text{Energy} = \text{Power} \times \text{Time}$
 - Output energy = Power station output $\times$ Number of seconds in one day

- Output energy = PS
- Total energy available per day = Number of trains per day $\times$ Mass of coal on a train $\times$ Energy from 1 kg of coal
- Total energy available per day = NMJ
- Efficiency $\frac{\text{output energy per day}}{\text{total energy available per day}} = \frac{PS}{NMJ}$

3

The correct answer is **A** because:

- Power is the rate of transfer of energy
 - $Power = \frac{\text{energy}}{\text{time}}$
- Water in the lake possesses gravitational potential energy
 - $Power = \frac{GPE}{\text{time}} = \frac{mgh}{t} = \left(\frac{m}{t}\right)gh$
- The mass flow rate of the water $\left(\frac{m}{t}\right)$ is 2400 kg min^{-1}
- This means that every minute, 2400 kg of water flows through the turbine
 - In 60 s, 2400 kg of water flows
 - Therefore, in 1 s, $\frac{2400}{60} = 40 \text{ kg}$ of water flows
- The mass flow rate is equal to $\frac{m}{t} = 40 \text{ kg s}^{-1}$
 - Power of water = $\left(\frac{m}{t}\right)gh = 40 \times 9.81 \times 90 = 35\,316 \text{ W}$
- This is the maximum power available from the turbine, however, the turbine has an efficiency of 75%
 - Power output of the turbine = $0.75 \times 35\,316 = 26\,487 = 26 \text{ kW}$

4

The correct answer is **B** because:

- To stack the blocks on top of each other, we only need to lift three of them, as the bottom block is already on the table
- When on the ground, a block has zero gravitational potential energy
- When at a height x above the ground, the gravitational potential energy of a

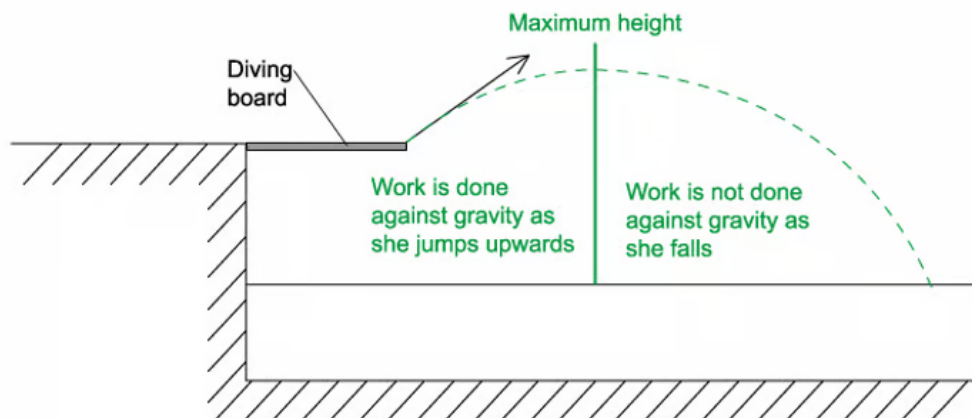
block is mgx

- So, the work that needs to be done to lift a block from the ground to a height x is mgx , where x will equal $h, 2h, 3h$
 - The first block is lifted by a height h , so, work done = mgh
 - The second block is lifted by a height $2h$, so, work done = $2mgh$
 - The third block is lifted by a height $3h$, so, work done = $3mgh$
- Total work done = $mgh + 2mgh + 3mgh = 6mgh$

5

The answer is **A** because:

- Work done = Force $\times$ Distance moved in direction of the force
- In situation A, a girl dives from a diving board into a swimming pool
- The direction of the gravitational force is downwards
- The motion can be modelled as a projectile, so she initially travels up before falling into the swimming pool
- Work is only done against gravity as a body moves upwards, work is **not** done against gravity as she falls



B is correct because a man pushes a car along a level road:

direction of (electric) force = horizontal ✓

direction of motion = horizontal ✓

C is correct because an electron is accelerated towards a positively-charged plate:

direction of (electric) force = towards positively charged plate ✓

direction of motion = electron moves towards the positively charged plate ✓

D is correct because a piston is pushed outwards as a gas expands:

direction of force = on piston (the gas in the atmosphere exerts force on the surface of the piston) ✓

direction of motion = piston moves outwards ✓

6

The correct answer is **B** because:

- Efficiency = $\frac{\text{useful output power}}{\text{total power input}}$
- We know the power output is = 150 kW = 150×10^3 W
- Power input = $20 \times 40 \text{ MJ} = 20 \times 40 \times 10^6$
- We must convert the fuel consumption into litres per second
 - In one hour, the fuel consumed is 20 litres
 - In one minute, the fuel consumed is $\frac{20}{60}$ litres
 - In one second, the fuel consumed is $\frac{20}{60 \times 60}$ litres
- So, power input = $\frac{20 \times 40 \times 10^6}{60 \times 60}$
- Efficiency = $\frac{\text{output power}}{\text{power input}} = \frac{150 \times 10^3}{\frac{20 \times 40 \times 10^6}{60 \times 60}} = \frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$

7

The correct answer is **D** because:

- The car does work equal to , this is the total energy
- Efficiency = $\frac{\text{useful energy}}{\text{total energy}}$

- We know the power output is $= 150 \text{ kW} = 150 \times 10^3 \text{ W}$
- Power input $= 20 \times 40 \text{ MJ} = 20 \times 40 \times 10^6$
- We must convert the fuel consumption into litres per second
 - In one hour, the fuel consumed is 20 litres
 - In one minute, the fuel consumed is $\frac{20}{60}$ litres
 - In one second, the fuel consumed is $\frac{20}{60 \times 60}$ litres
- So, power input $= \frac{20 \times 40 \times 10^6}{60 \times 60}$
- Efficiency $= \frac{\text{output power}}{\text{power input}} = \frac{150 \times 10^3}{\frac{20 \times 40 \times 10^6}{60 \times 60}} = \frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$

7

The correct answer is **D** because:

- The car does work equal to , this is the total energy
- Efficiency $= \frac{\text{useful energy}}{\text{total energy}}$
- For constant velocity, the resultant force is zero. So, the force required to move the car up the slope should be equal to the component of weight along the slope
 - Component of weight against $F = mg \sin \alpha$
 - Work done to move car a distance $s = (mg \sin \alpha)s$
- Efficiency $= \frac{\text{useful energy}}{\text{total energy}} = \frac{(mg \sin \alpha)s}{Fs} = \frac{mg \sin \alpha}{F}$

8

The correct answer is **C** because:

- Work done = Force applied $\times$ Distance moved in the direction of force
- In this case the force acting on the mass is the tension in the string holding the mass which is equal to the weight W in this case
- The distance moved by the weight in direction of this force is q
 - So, work done $= Wq$

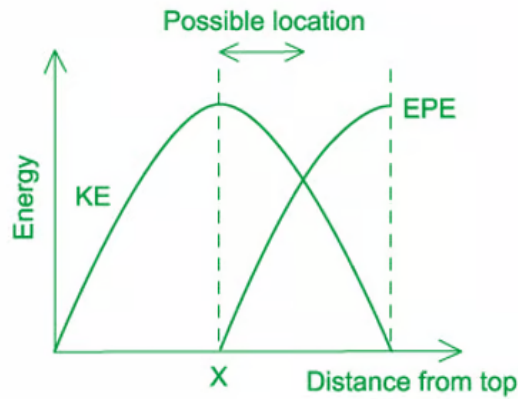
A & D are incorrect because if the force had also been given along the track then the work done could be represented as F_p or F_r or , but since we do know the force in the vertical direction is weight W , then we can only specify the work done by the weight

B is incorrect because these distances are in different directions, whereas work done requires the distance to be in the same direction as the force

9

The correct answer is **D** because:

- From the conservation of energy, the sum of energies halfway and at the bottom of the jump should be equal to the initial amount of gravitational potential energy the student possesses at the beginning of the jump
- When the jumper is halfway down, the rope will unwind fully but it will not have stretched yet. So, it will have a small elastic potential energy
- The kinetic energy will reach a maximum halfway down
- The graph below might give you a clearer image
- Point X shows the point where the jumper starts to be restricted by the pull of the rope
- Before X, the jumper undergoes free fall and hence the speed and kinetic energy increases
- After X, the elastic potential energy of the elastic rope starts increasing – this corresponds to a drop in the kinetic energy of the jumper, according to the principle of conservation of energy
- When the jumper is halfway down, their possible location is shown on the graph, it can clearly be seen that the kinetic energy is greater than the elastic potential energy at this point, making D the correct answer



10

The correct answer is **D** because:

- Efficiency = $\frac{\text{power output}}{\text{power input}}$
- Power output = IV
- Power output = $32 \times 230 = 7360$
- Water in the top section possesses gravitational potential energy
 - $\text{Power} = \frac{\text{GPE}}{\text{time}} = \frac{mgh}{t} = \left(\frac{m}{t}\right)gh$
- The mass flow rate is equal to $\frac{m}{t} = 200 \text{ kg s}^{-1}$
 - Power input of water = $\left(\frac{m}{t}\right)gh = 200 \times 9.81 \times 8 = 15\,696 \text{ W}$
- % Efficiency = $\frac{\text{power output}}{\text{power input}} \times 100\% = \frac{7360}{15\,696} \times 100 = 46.9\%$
- Therefore, the efficiency of the system is 47%

11

The correct answer is **D** because

- Power is equal to = $\frac{\text{energy}}{\text{time}} = \frac{Fd}{t} = Fv$

- Force is defined as the rate of change of momentum, so, we first need to find an expression for force in order to find the power
- sand drops vertically on to the belt at a rate of $m \text{ kg s}^{-1}$
 - Therefore, the rate at which the sand falls is $= \frac{\Delta \text{mass}}{\text{time}} = \left(\frac{\Delta m}{t} \right) = m \text{ kg s}^{-1}$
- Next, consider the momentum : $p = Mv$
- Rate of change of momentum $= \frac{\Delta p}{t} = \frac{Mv}{t} = \left(\frac{\Delta M}{t} \right) v = mv$
 - Therefore, in one second, the momentum of mass m of sand increases from Zero to mv
- Recall the force is defined as the rate of change of momentum, so the force involved is mv
- Force = rate of change of momentum $= mv$
 - so, Power $= Fv = mv \times v = mv^2$

12

The Correct answer is C because

- We are told the resistive force F is proportional to v^2
 - $F \propto v^2$
 - $F = kv^2$ where k is constant
- When $F=800 \text{ N}$, $V = 20 \text{ m s}^{-1}$
 - $k = \frac{F}{v^2} = \frac{800}{(20)^2} = 2$
- so, $F = 2v^2$
- Power output is equal to $P = Fv$
- Since $F = 2v^2$ we have :
 - Output power $P = Fv = (2v^2) \times v = 2v^3$
- So, when $V = 40 \text{ m s}^{-1}$
 - Output power $P = 2v^3 = 2 \times (40)^3 = 128\,000 \text{ W}$
- Therefore, the effective power output is 128 kW

13

- Power output = IV
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- Water in the top section possesses gravitational potential energy
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- Since $F = 2v^2$ we have :
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- So, when $v = 40 \text{ m s}^{-1}$
 - Output power $P = 2v^3 = 2 \times (40)^3 = 128\,000 \text{ W}$
- Therefore, the effective power output is 128 kW

13

The correct answer is **D** because :

- From the question we are told :
 - Increase in gravitational potential energy $= E_p$
 - Increase in kinetic energy $= E_k$
 - Time taken $= t$
- Let the total gravitational energy be equal to E
- We know that gravitational energy is the sum of potential energy and kinetic energy so
 - $E = E_p + E_k$
- Let P be the power input of the mass
- Power is equal to $P = \frac{E}{t}$
 - Therefore : $P = \frac{E}{t} = \frac{E_p + E_k}{t}$

